

PBA Array Warpage Analysis Report

Analysis Folder: Analysis Data

Number of Files: 10

Date: 2025-09-29 14:41:15

PBA Warpage analysis

Main Analysis Contents:

- Individual PBA array heatmap analysis
 - Statistical comparison analysis across arrays
 - 3D surface visualization of warpage patterns
 - Advanced statistical analysis
-

Generated by Warpage Analysis Tool v2.0
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Legend & Terminology

STATISTICAL METRICS

Basic Statistics:

- Mean: Average warpage value across all measurement points on PBA array
- Standard Deviation (Std): Measure of data spread around the mean
- Range: Difference between maximum and minimum warpage values (Max - Min)
- Min/Max: Minimum and maximum warpage values in the PBA array dataset

Advanced Analysis:

- Hotspots: Areas on PBA array where warpage exceeds 95th percentile threshold
- Local Variability: Spatial variation in warpage across the PBA surface
- Gradient Magnitude: Rate of change in warpage values across the array
- Contour Plots: Lines connecting points of equal warpage values
- CDF Plot: Cumulative distribution of (Max-Min) ranges across PBA arrays
- Correlation Matrix: Shows relationships between different PBA arrays

Distribution Analysis:

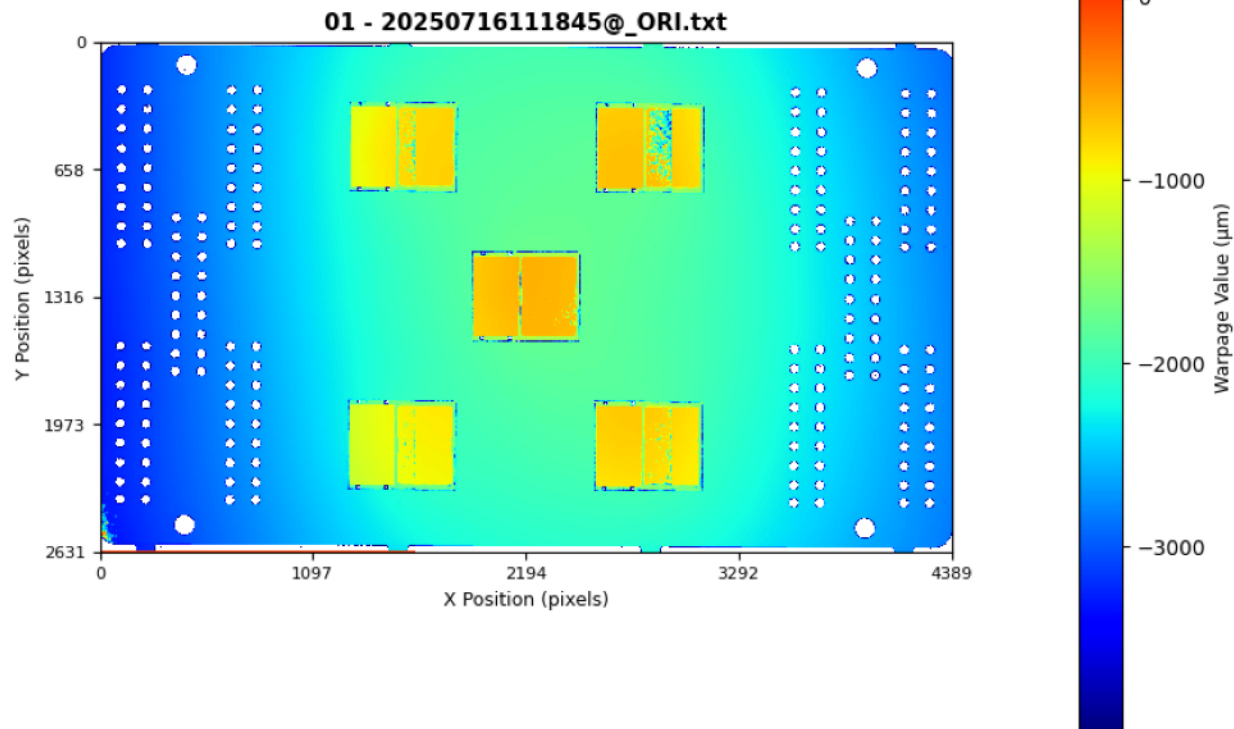
- Violin Plots: Extension of box plots showing probability density of data distribution
- Percentile Analysis: Analysis of percentile distribution in data

Generated by Warpage Analysis Tool v2.0

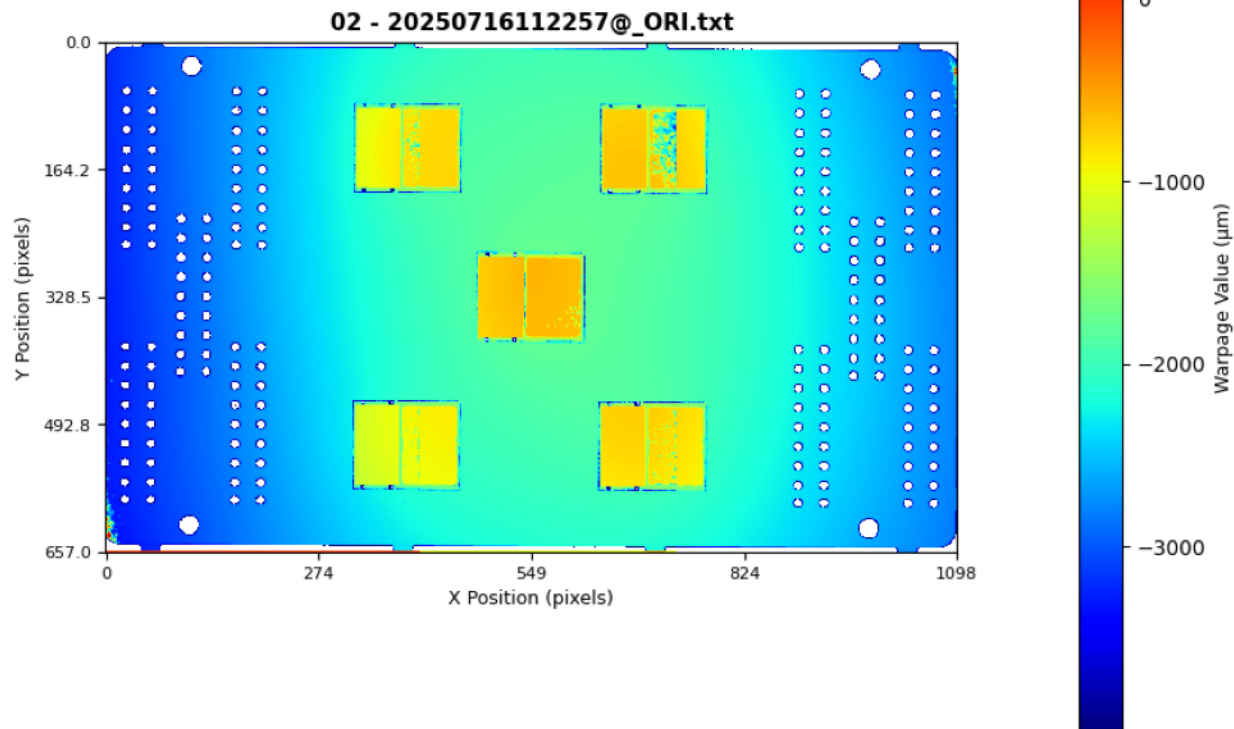
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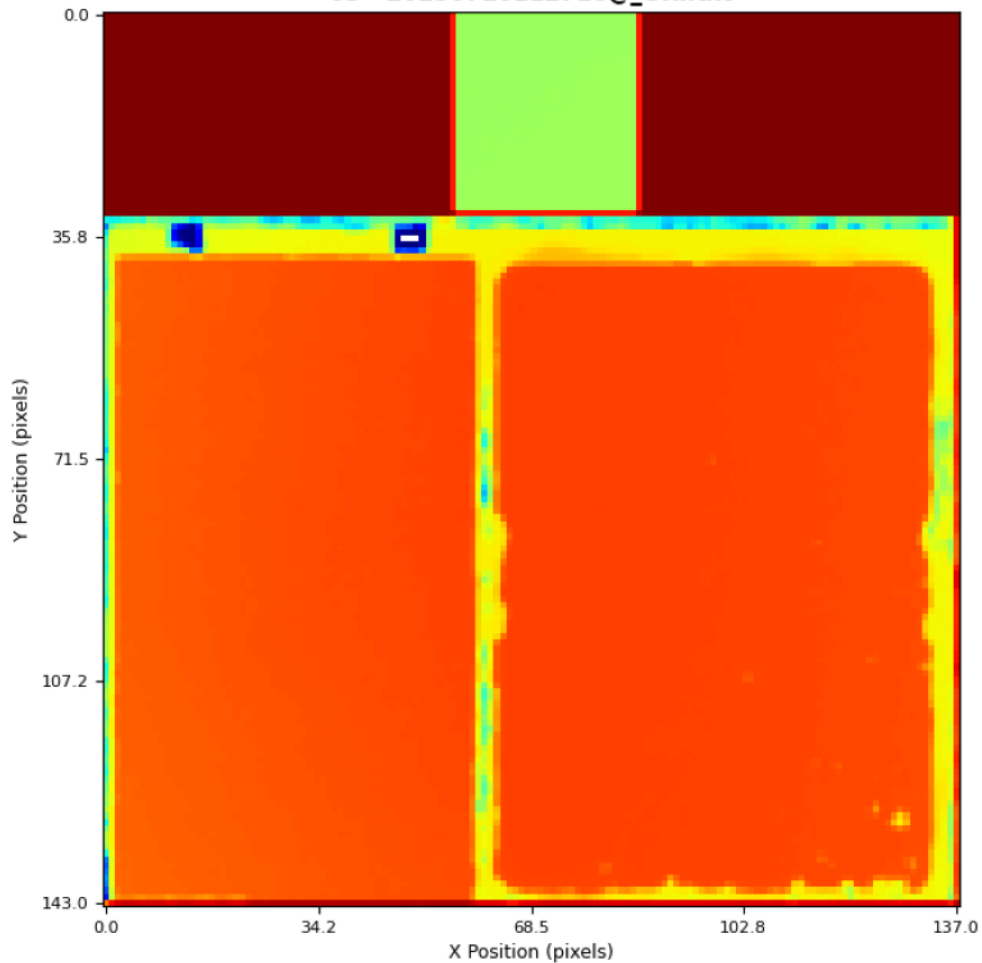
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Std: 628.035



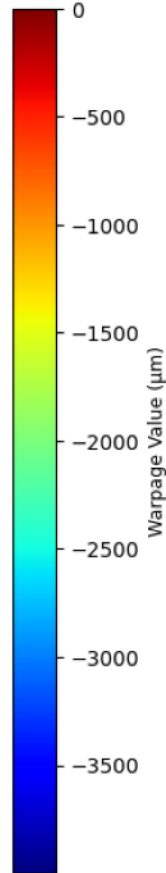
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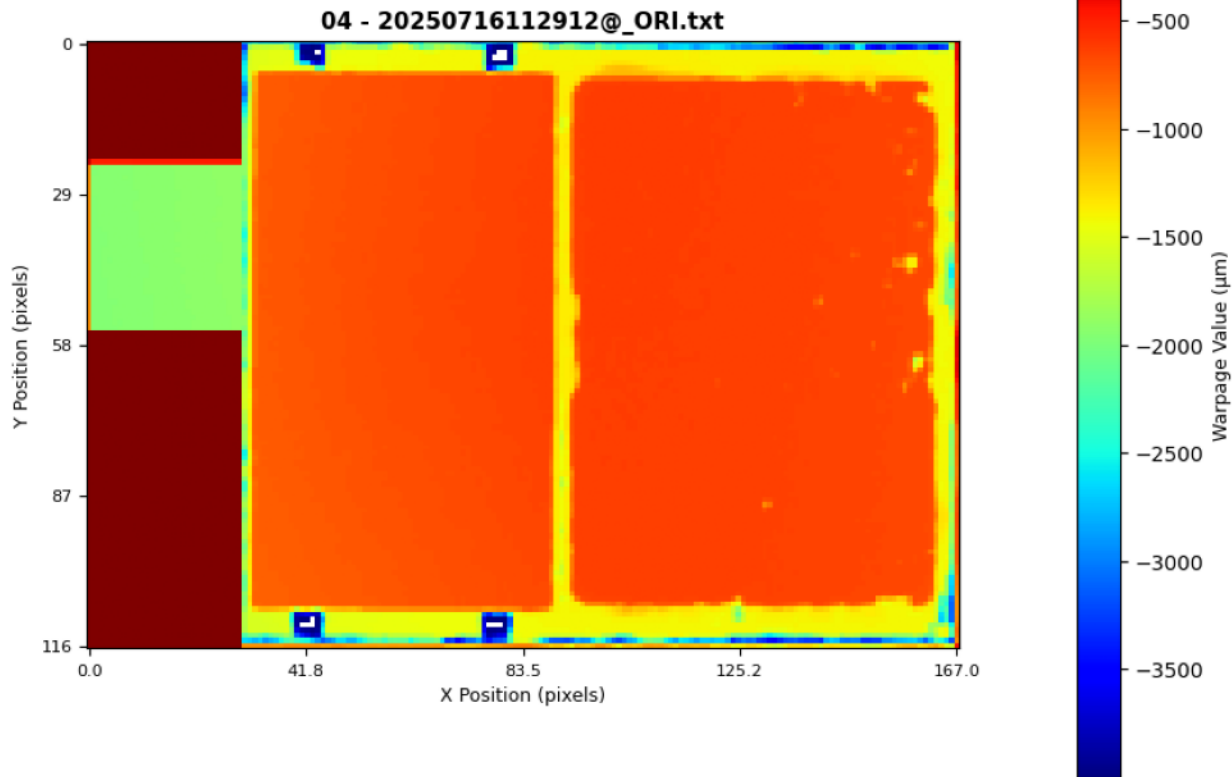
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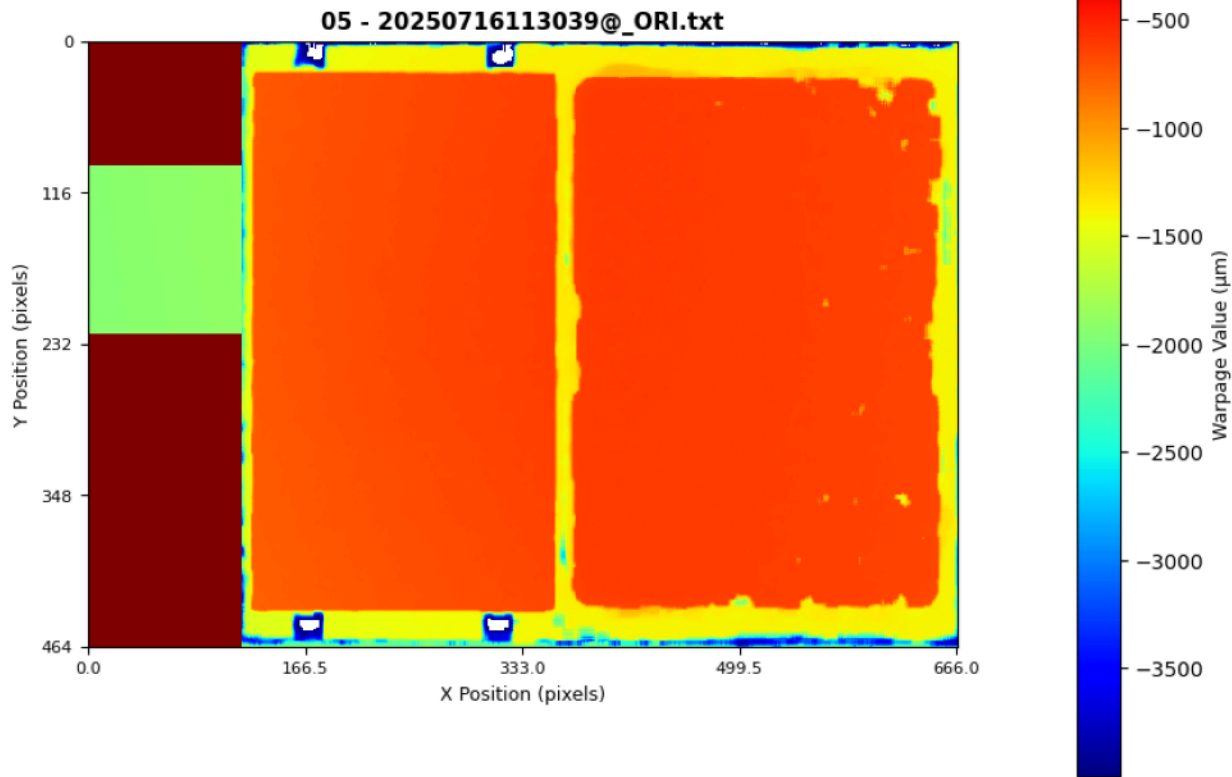
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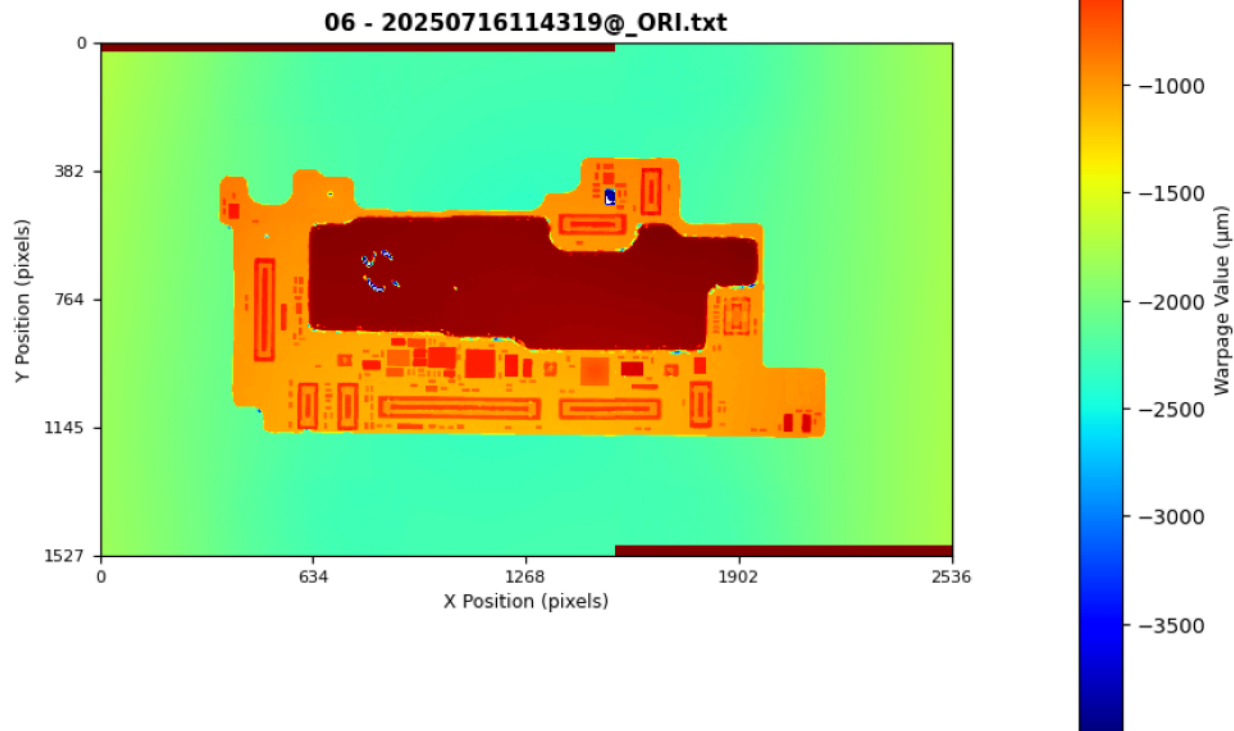
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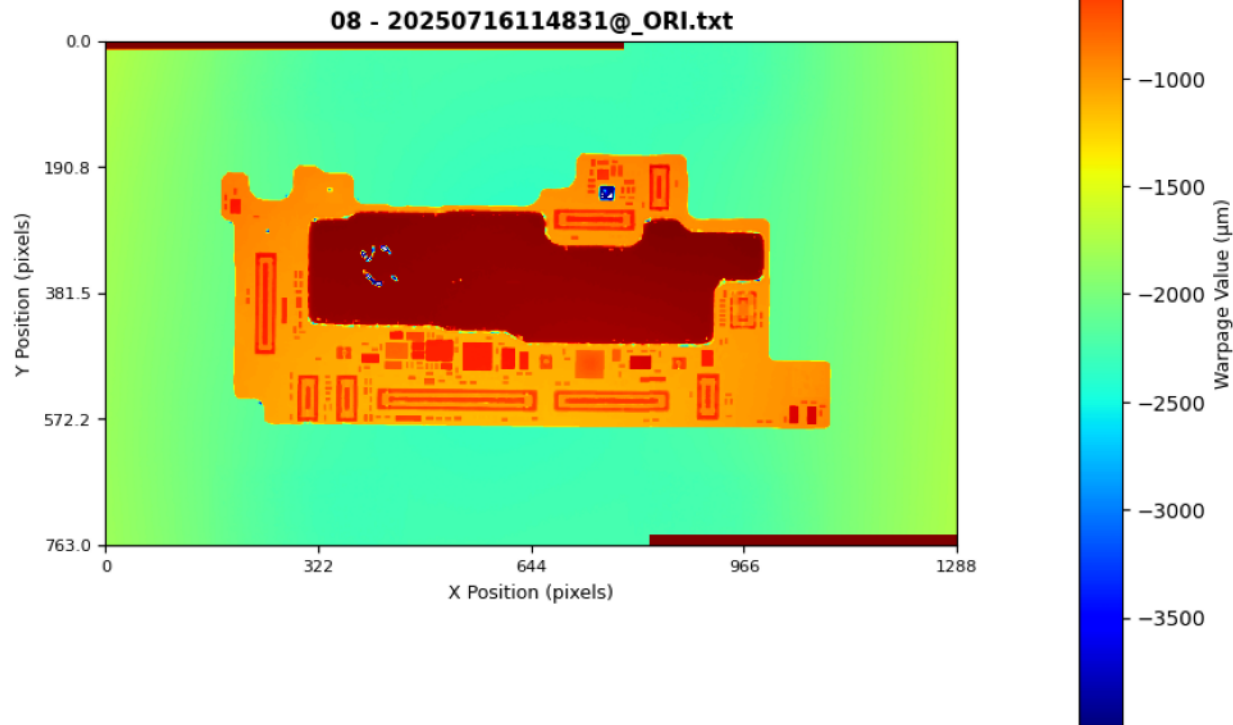
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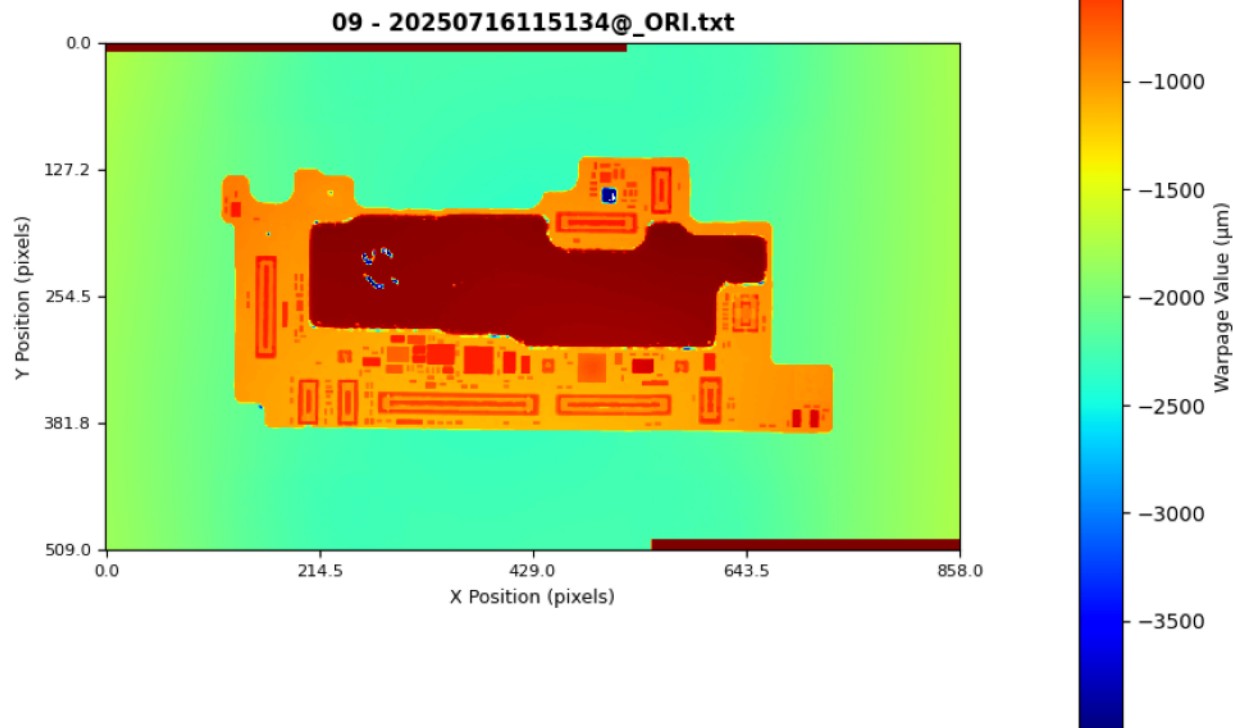
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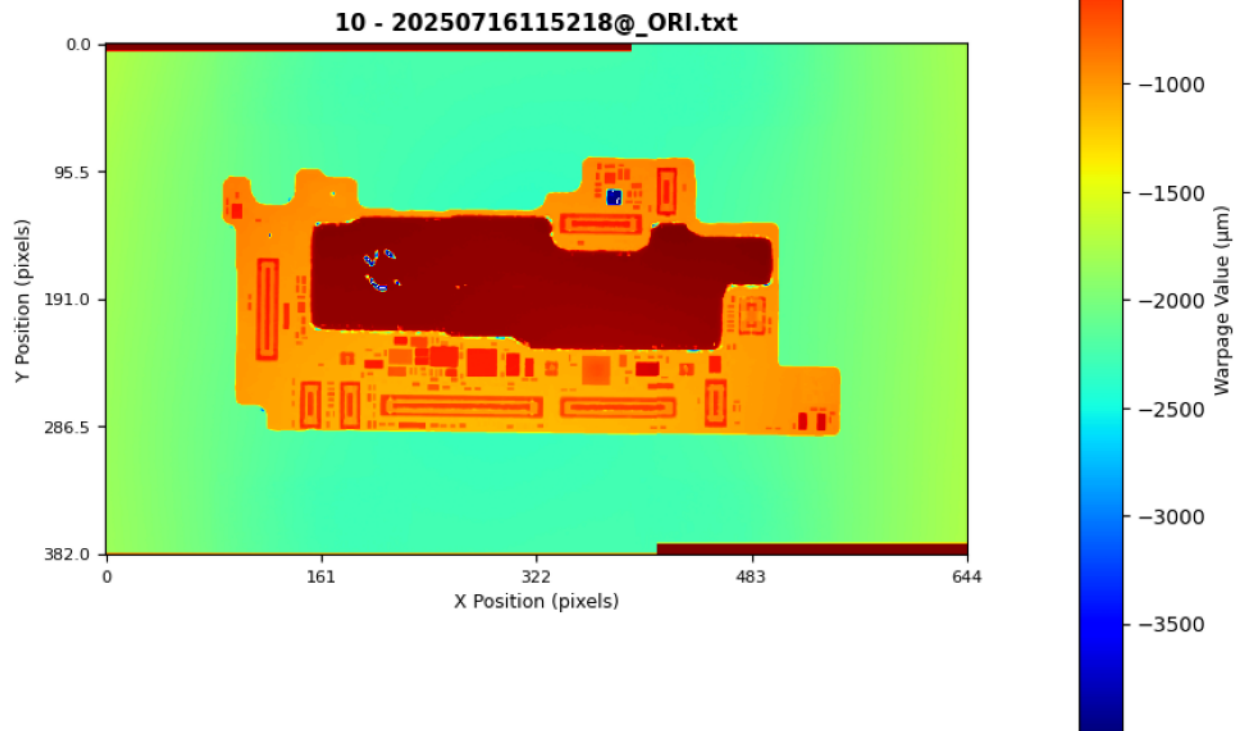
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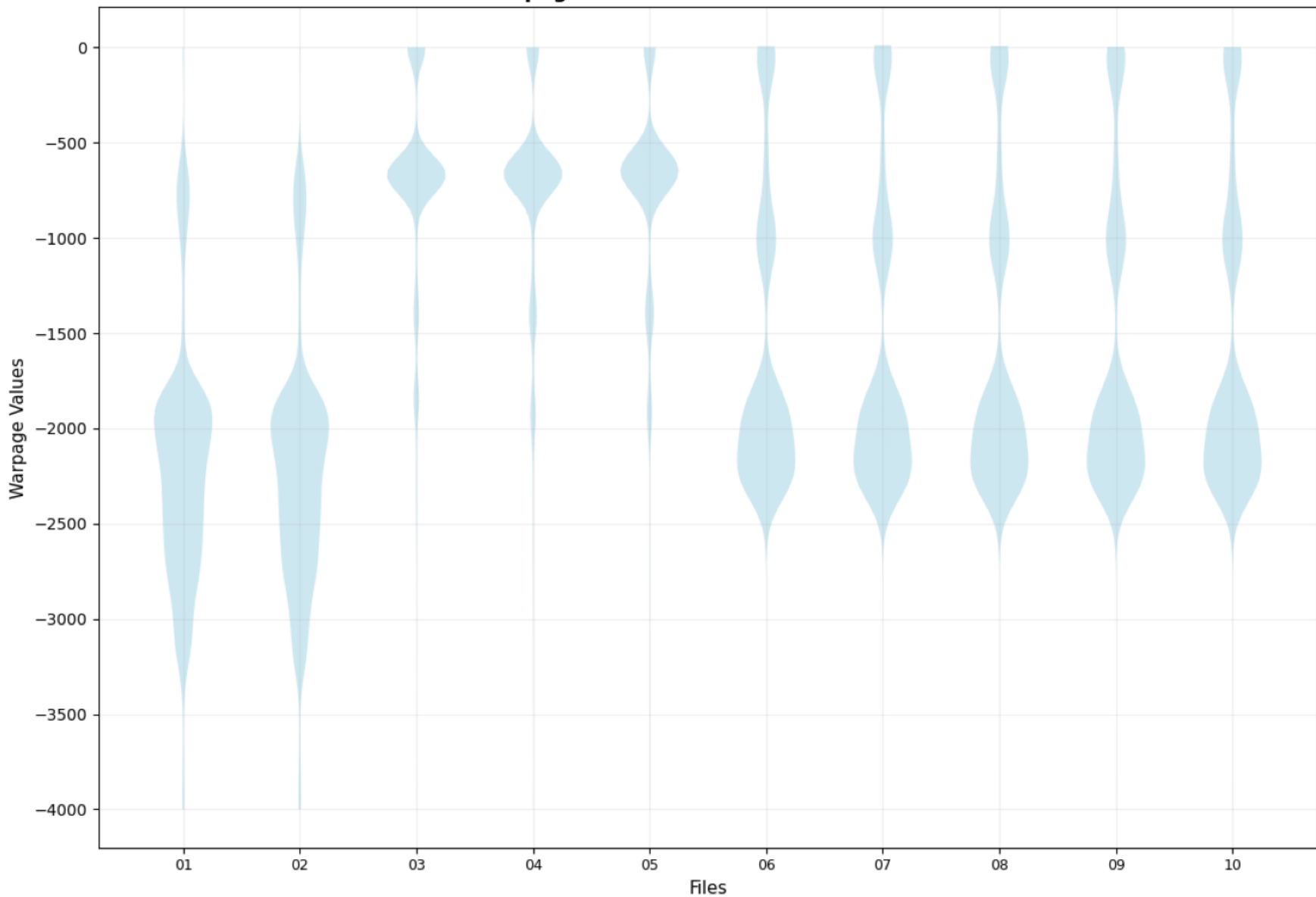
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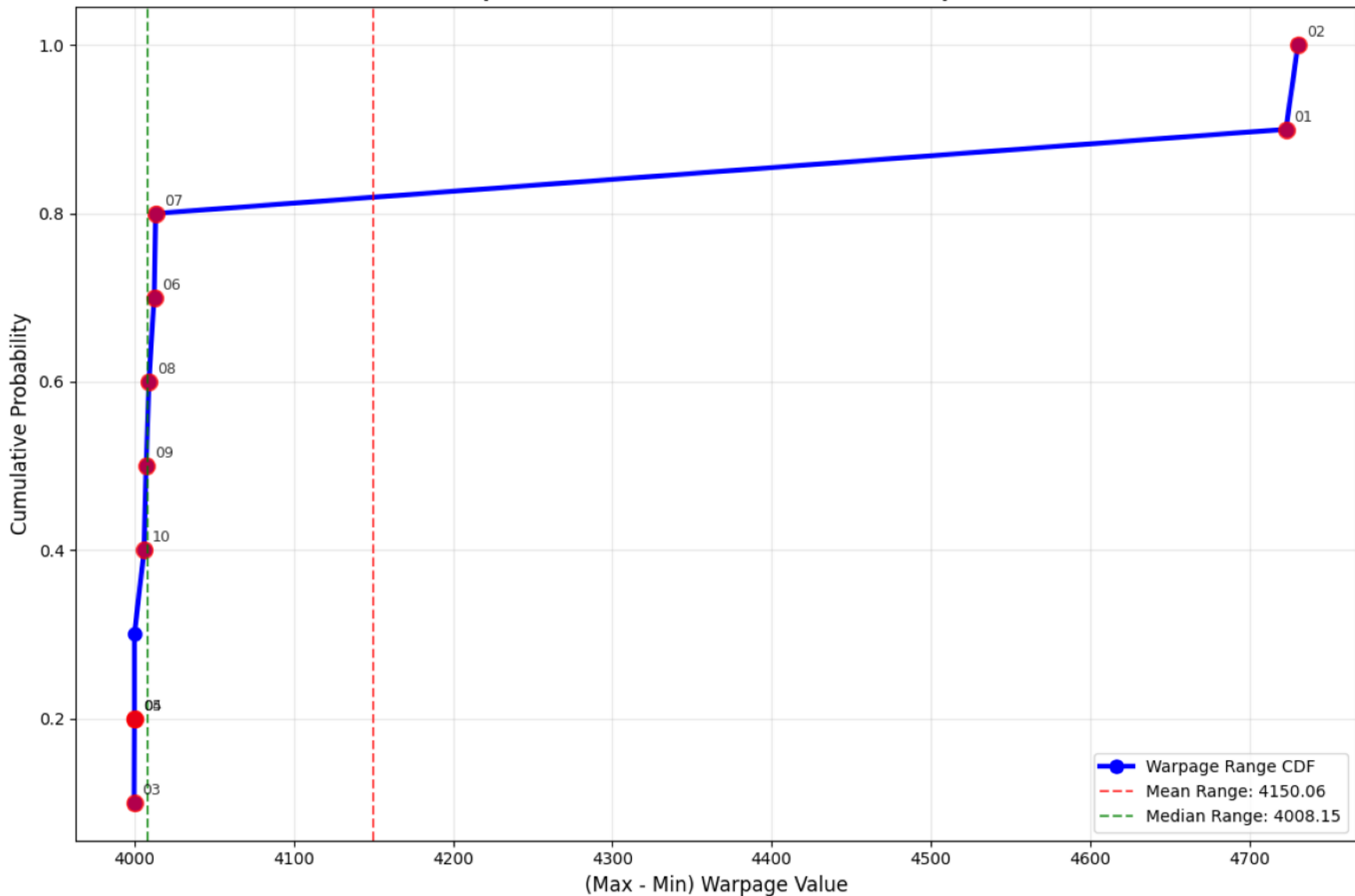
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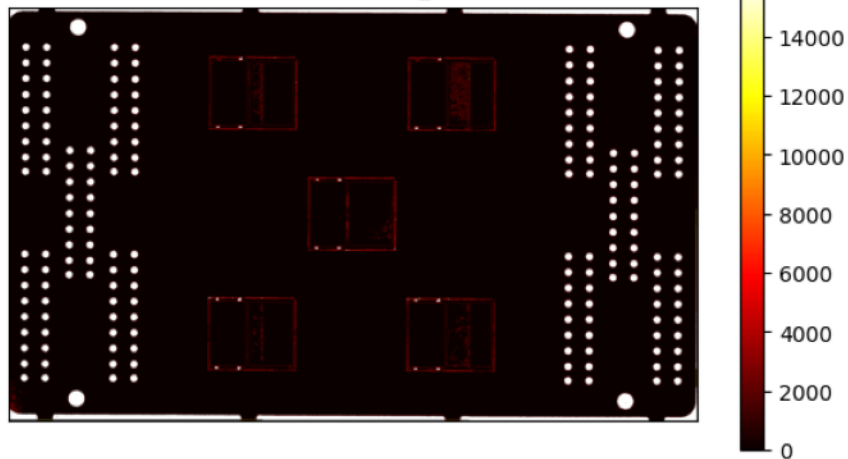
Warpage Distribution - Fast Violin Plots



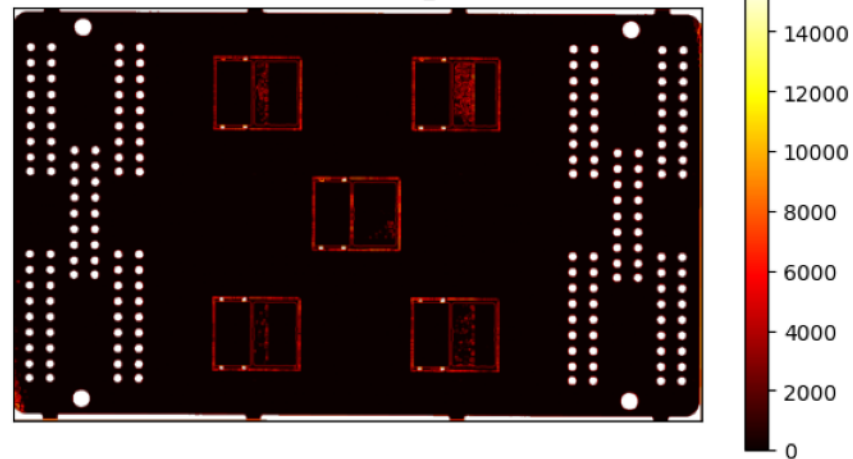
Cumulative Distribution of Warpage Ranges
(X-axis: Max-Min values across files)



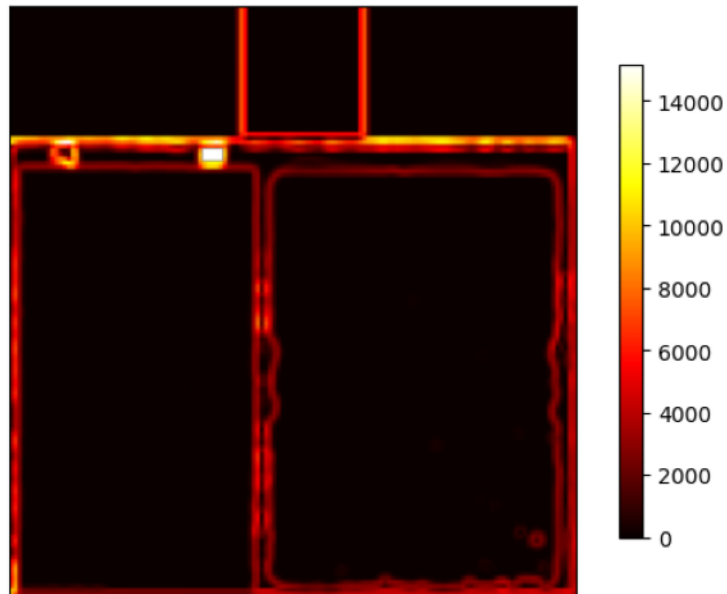
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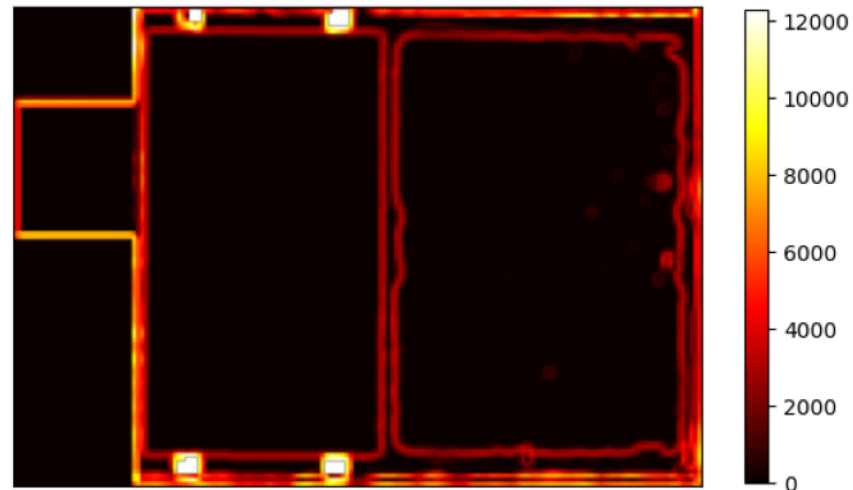
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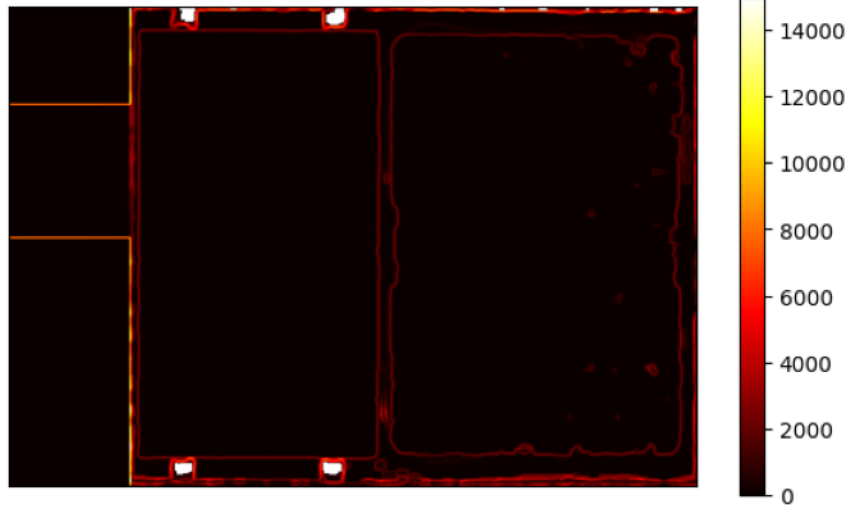
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04 - Gradient Magnitude
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05 - Gradient Magnitude
20250716113039@_ORI.txt



06 - Gradient Magnitude
20250716114319@_ORI.txt



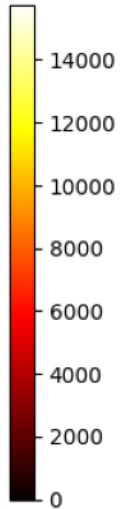
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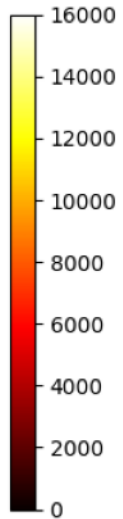
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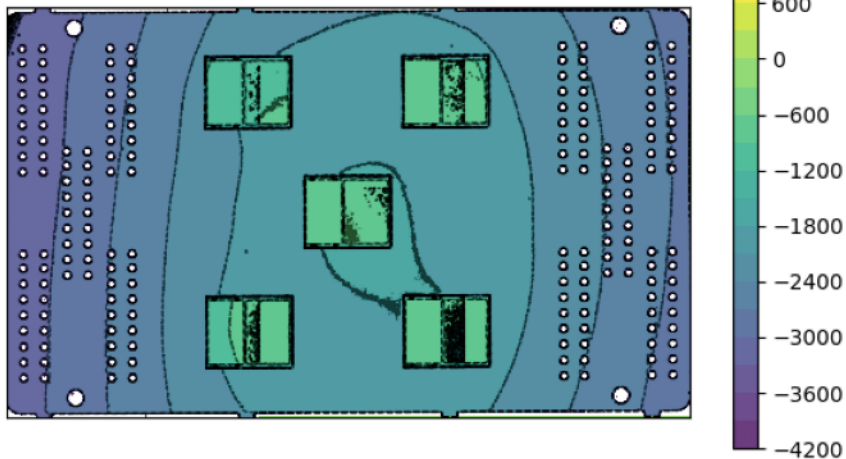
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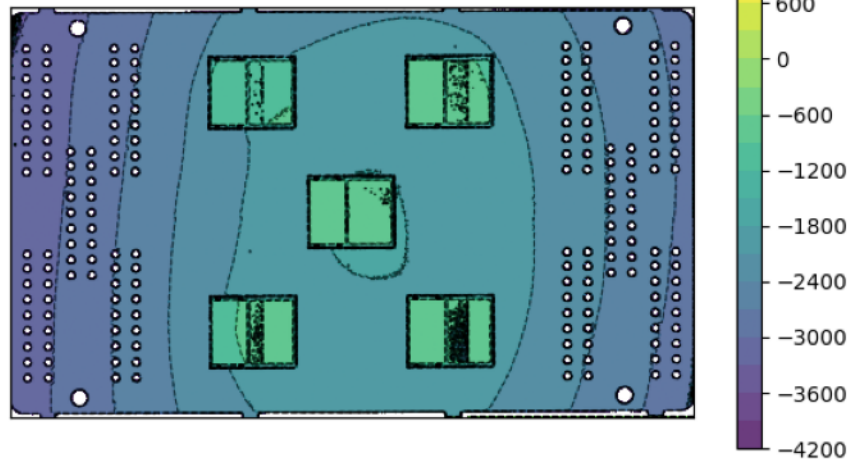
10 - Gradient Magnitude
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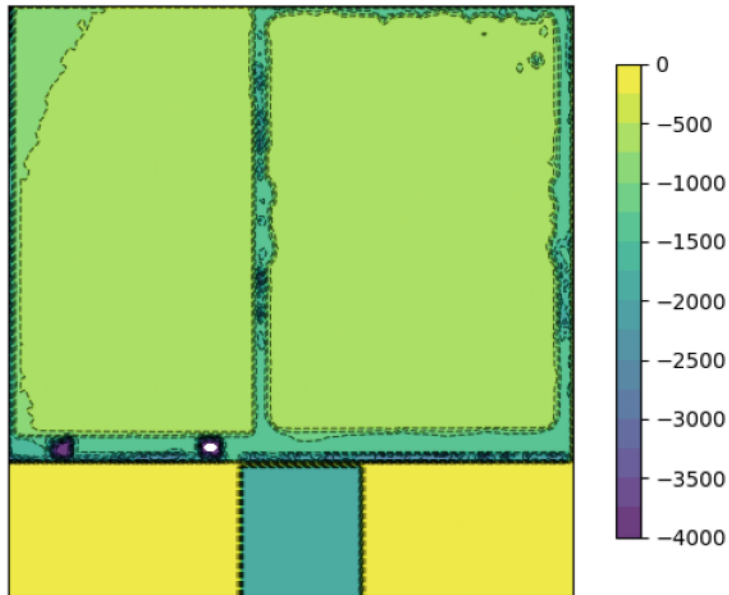
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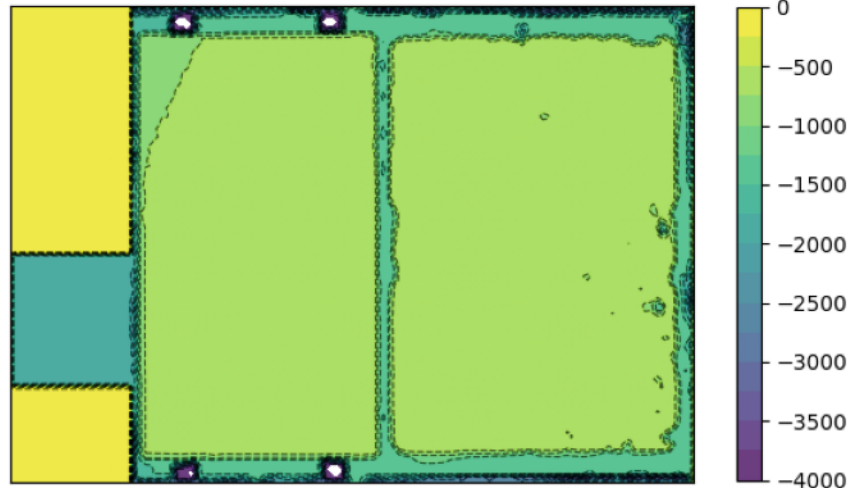
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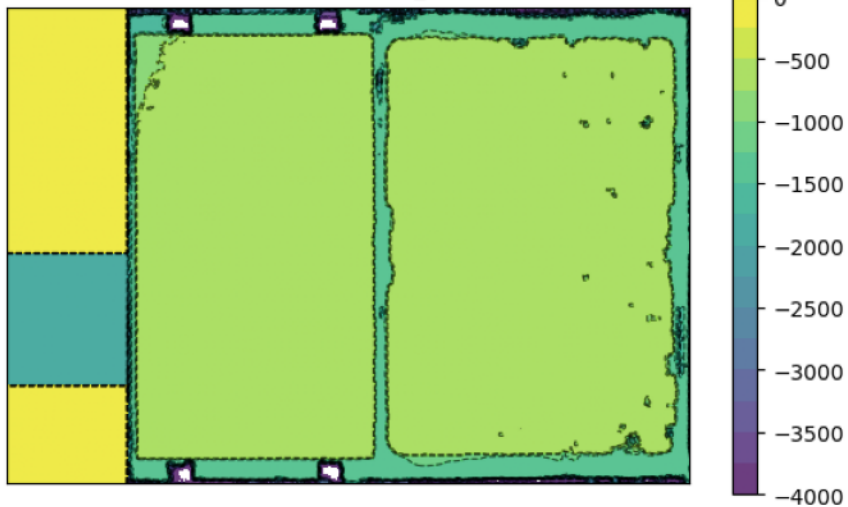
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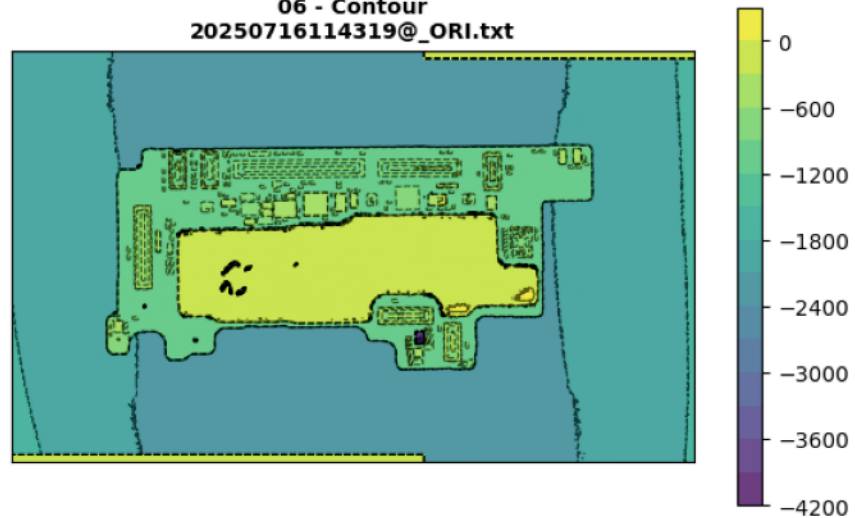
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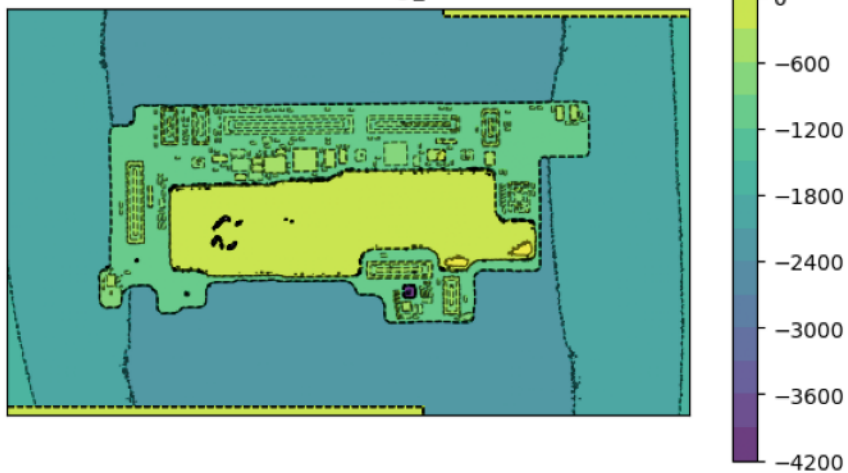
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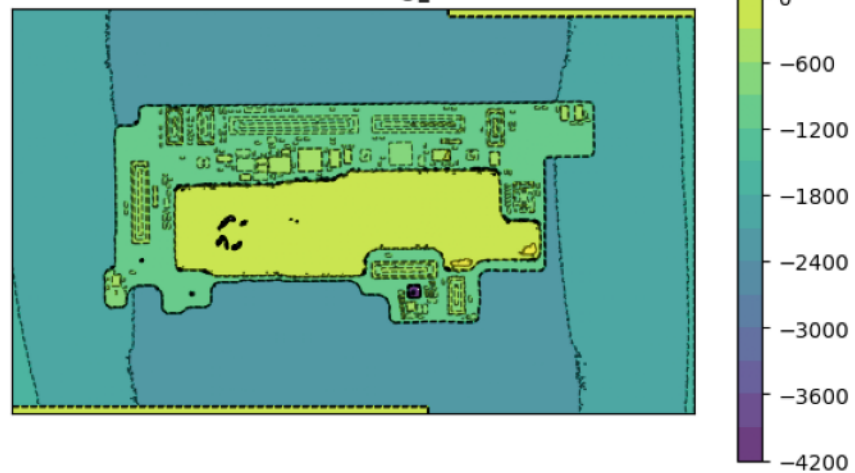
06 - Contour
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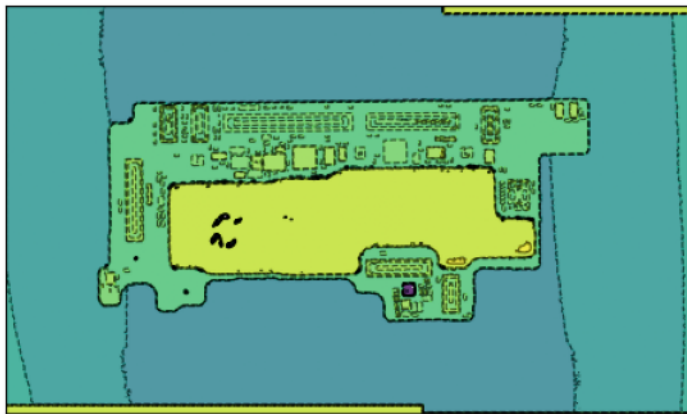
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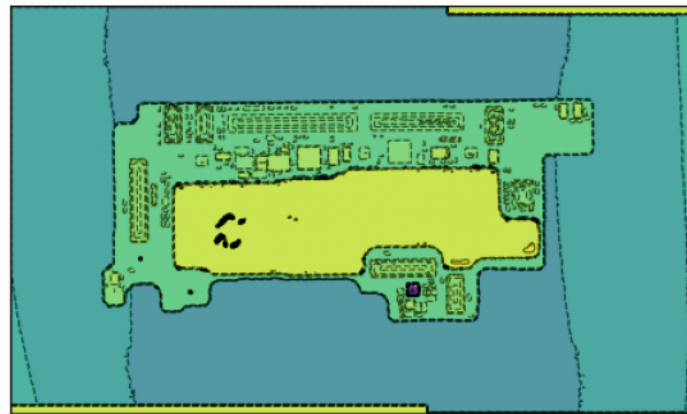
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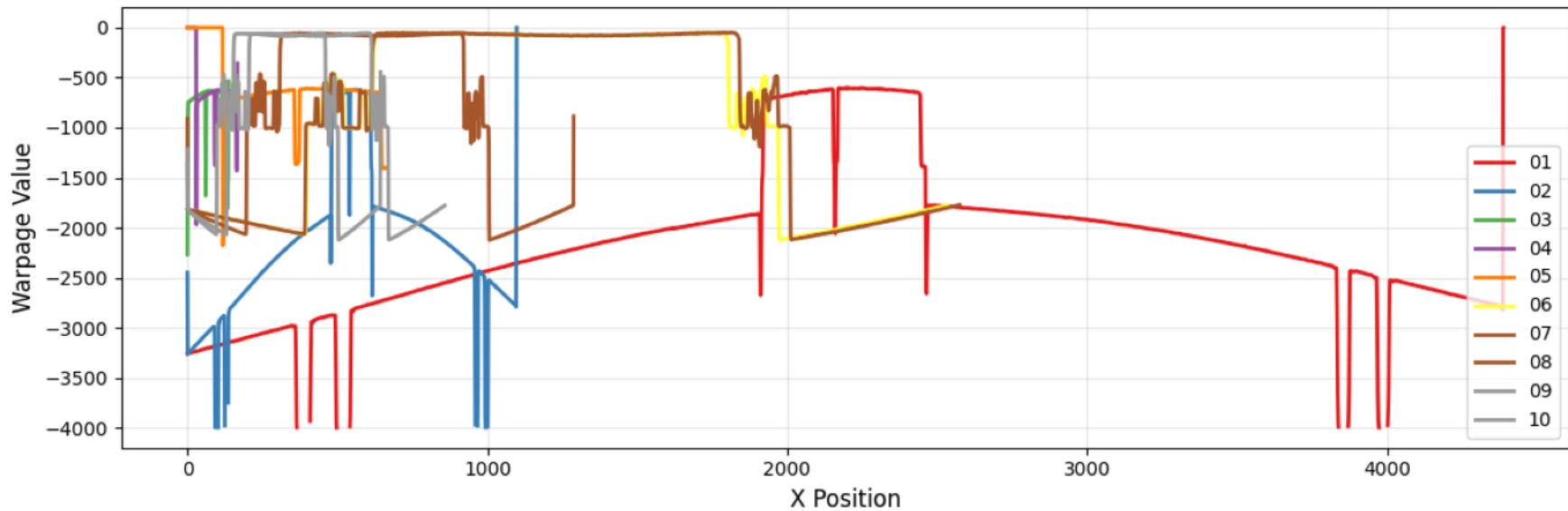
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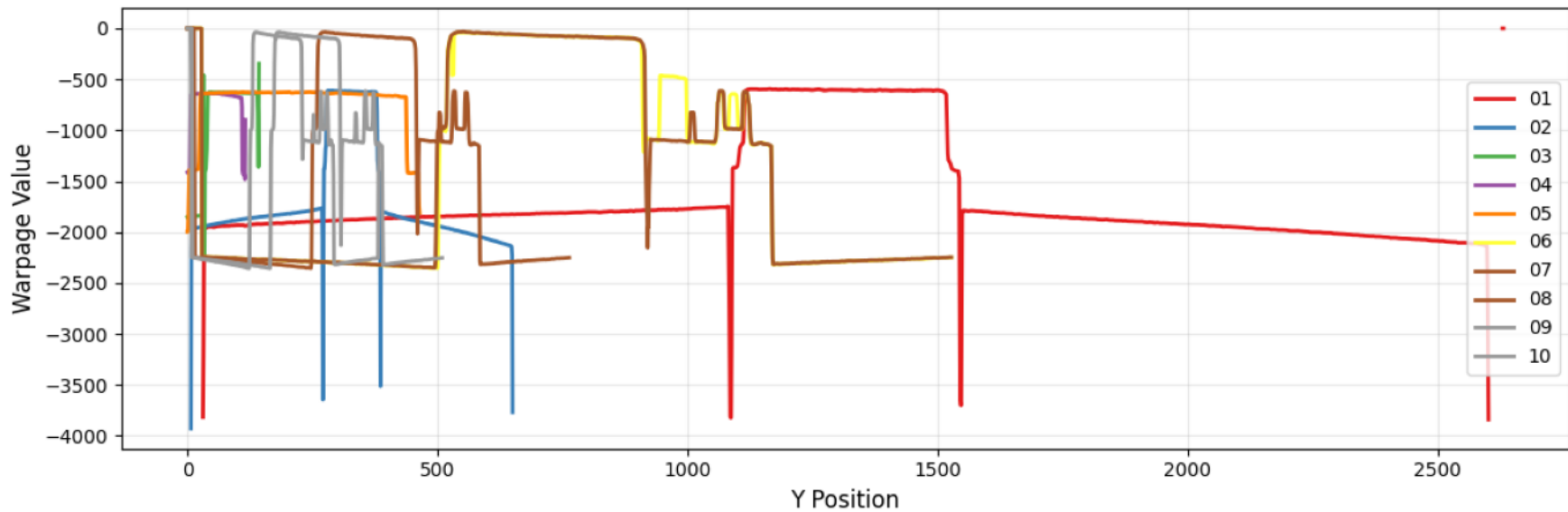
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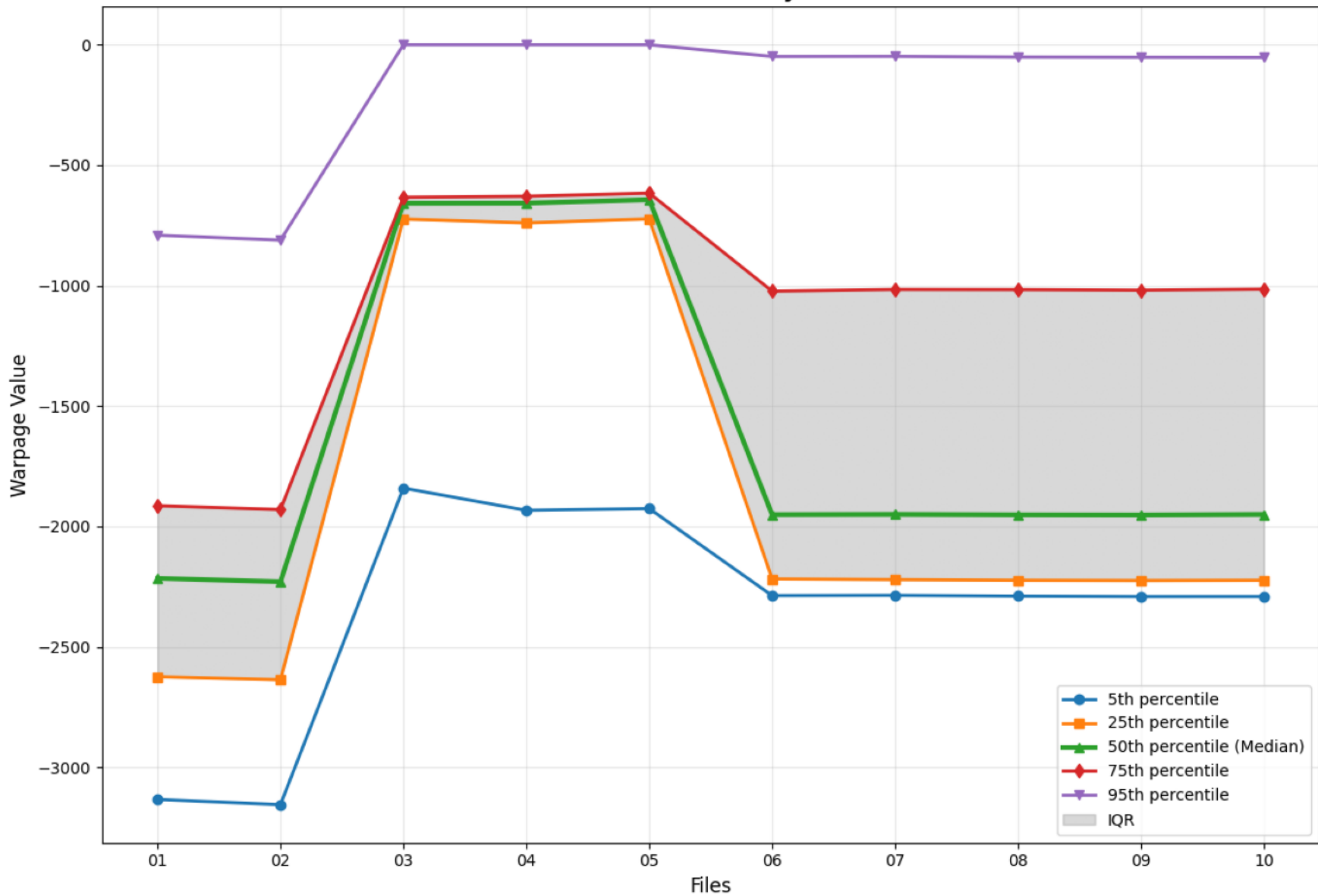
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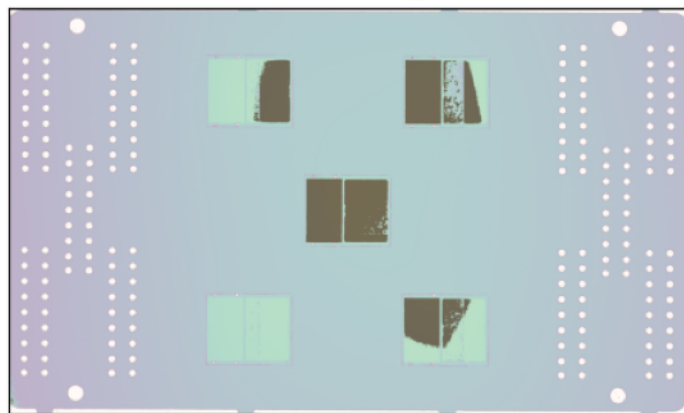
Center Column Profile



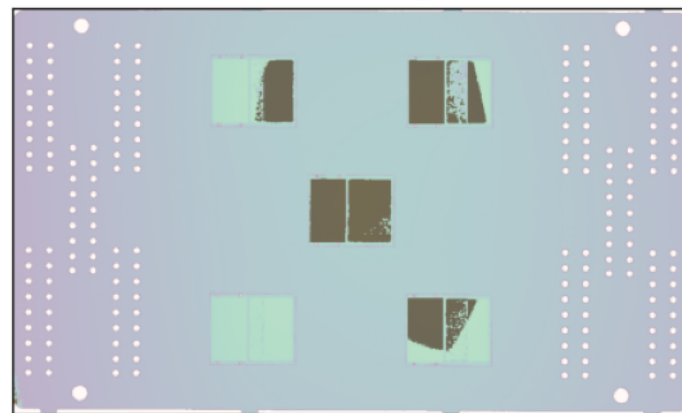
Percentile Analysis



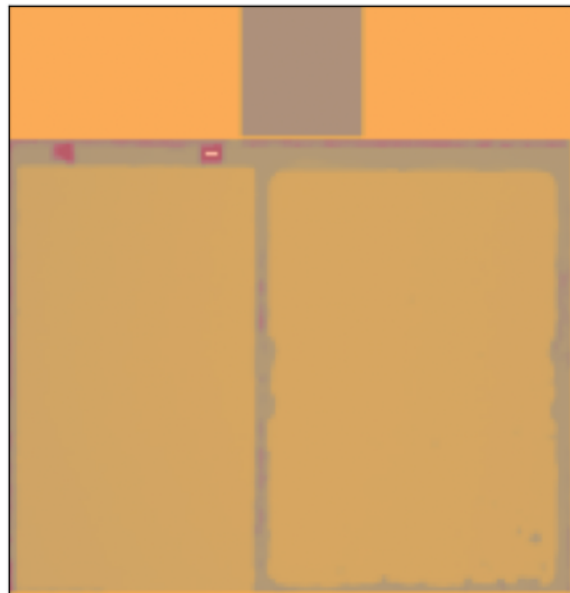
01 - Hotspots (>-790.5)
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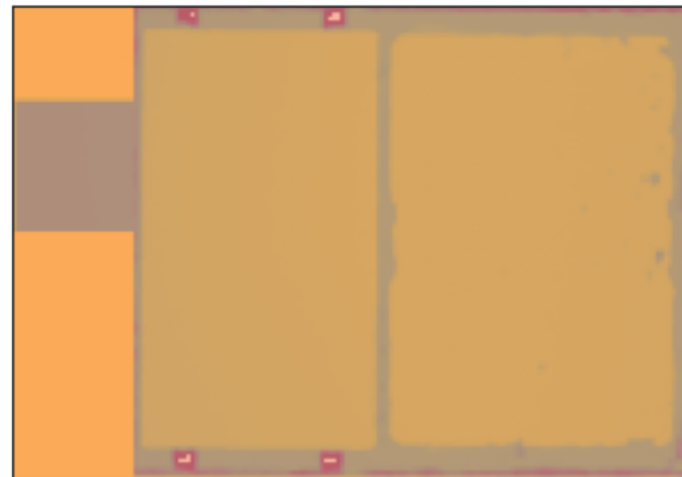
02 - Hotspots (>-811.1)
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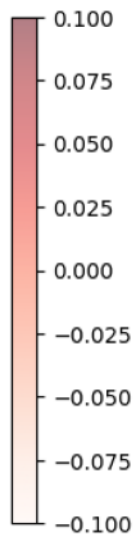
03 - Hotspots (>0.0)
20250716112716@_ORI.txt



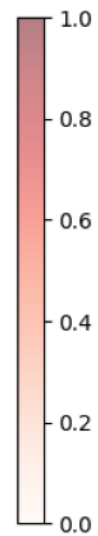
04 - Hotspots (>0.0)
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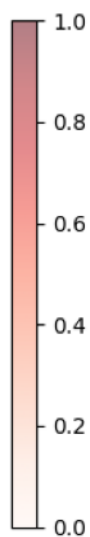
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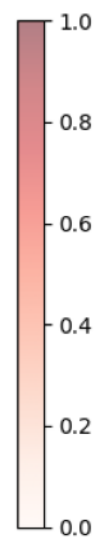
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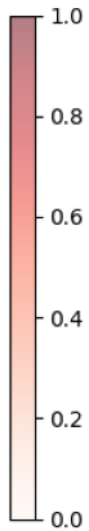
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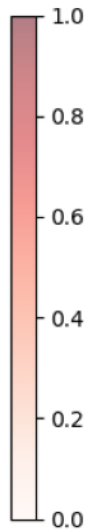
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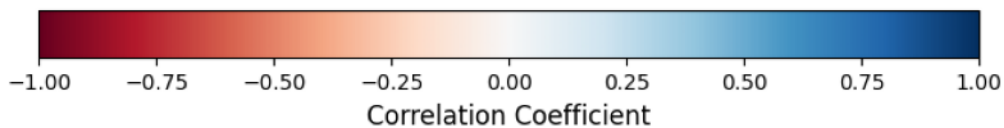
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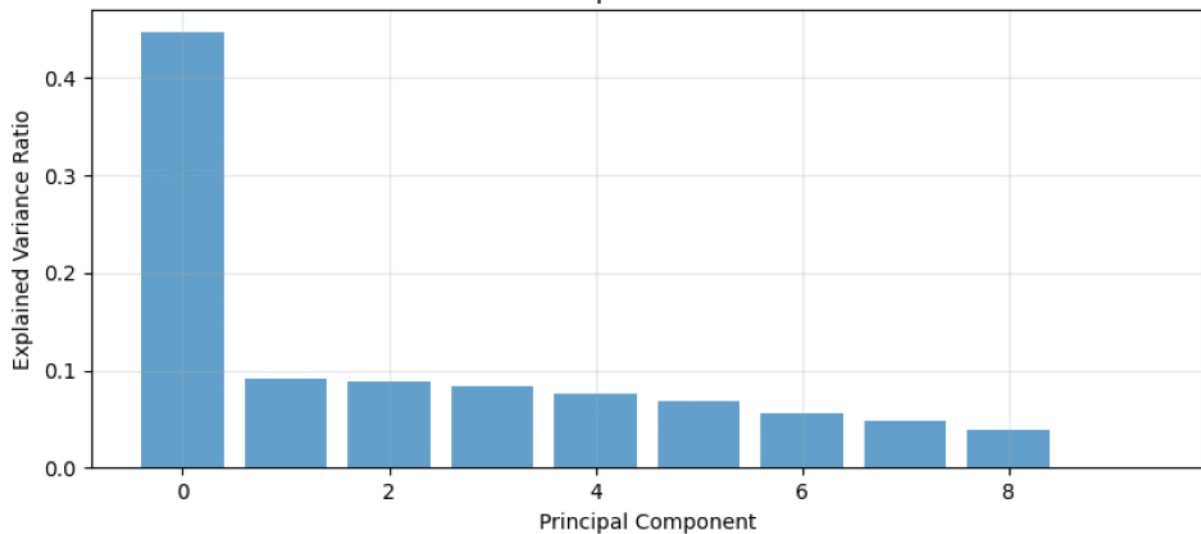
10 - Hotspots (>-52.7)
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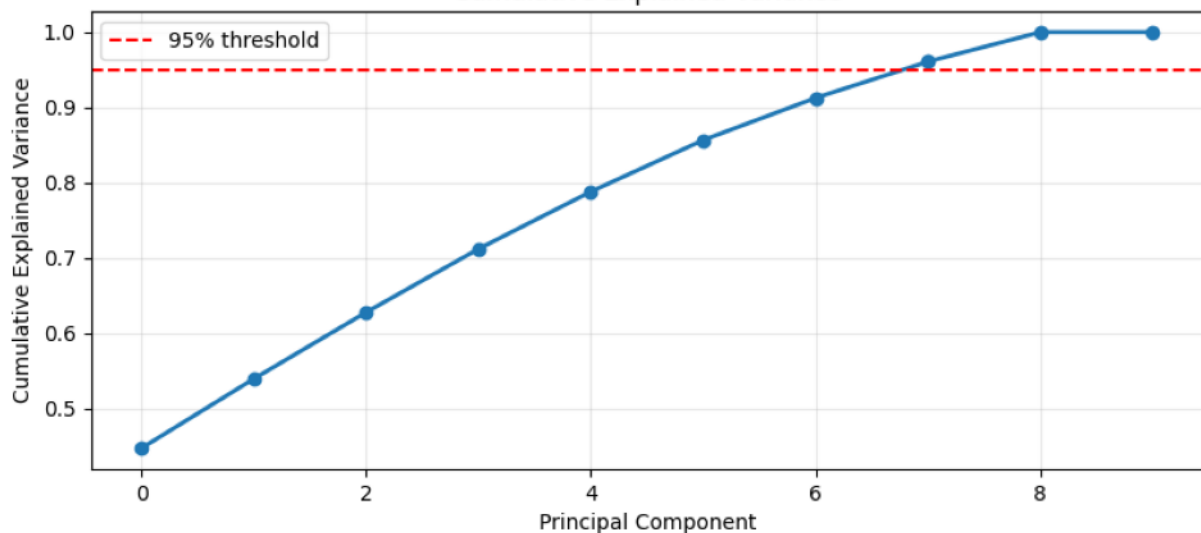
Correlation Matrix Between Files



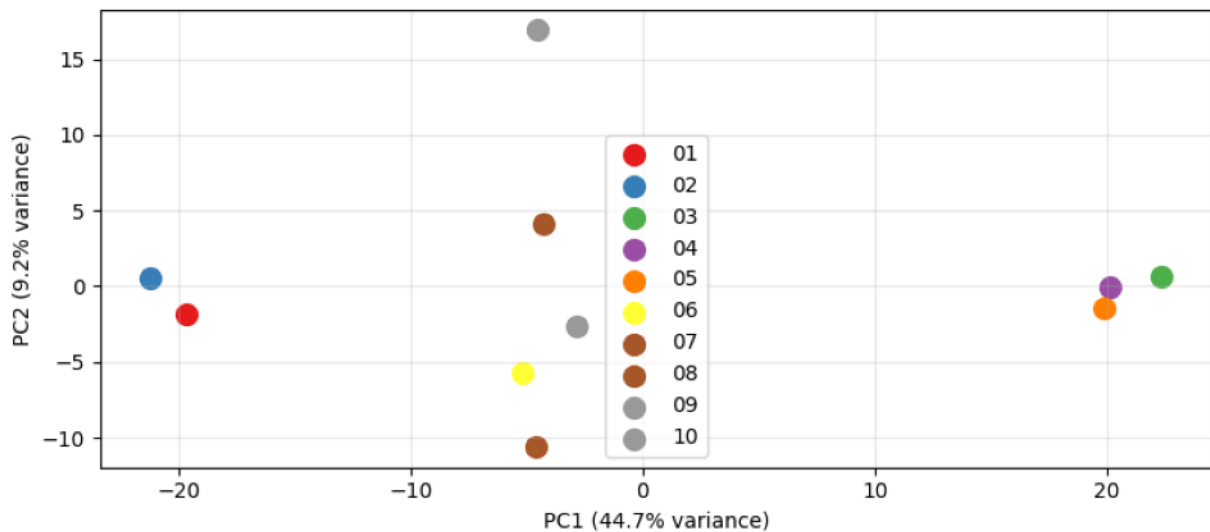
PCA - Explained Variance



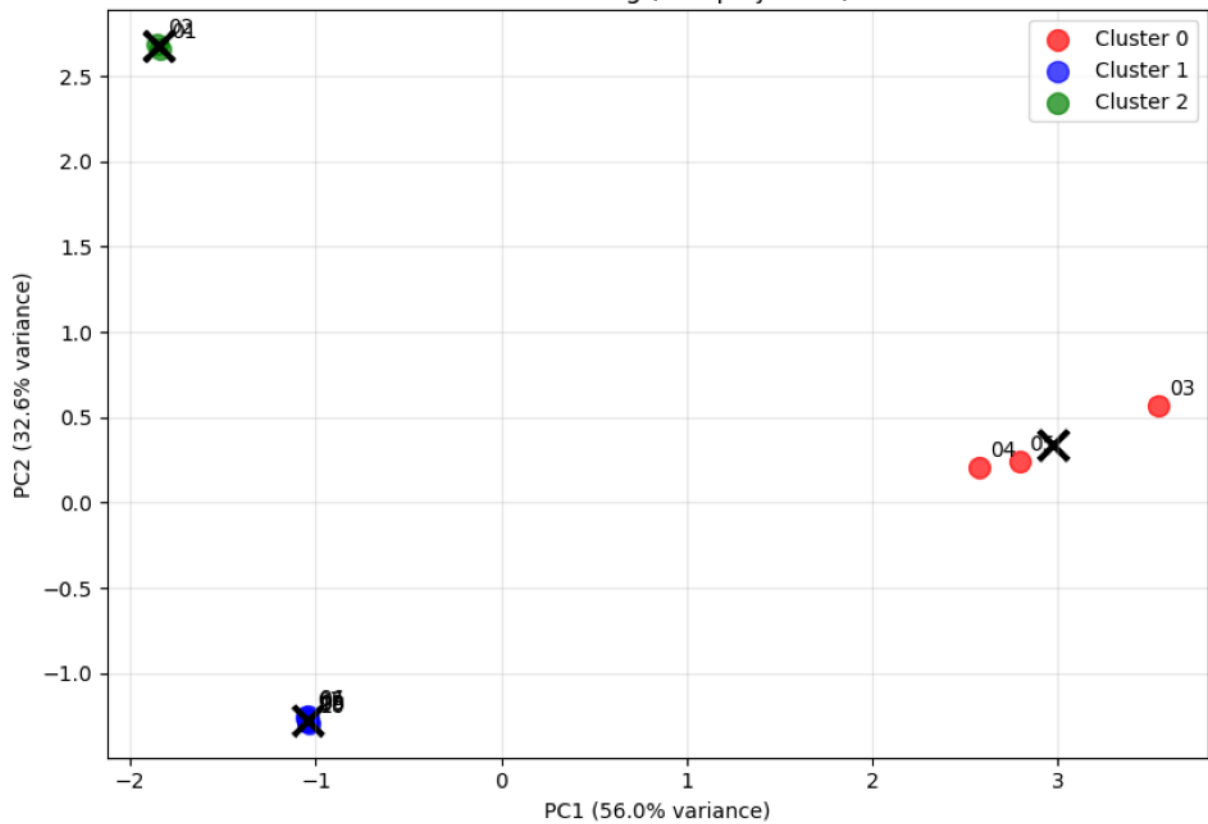
Cumulative Explained Variance



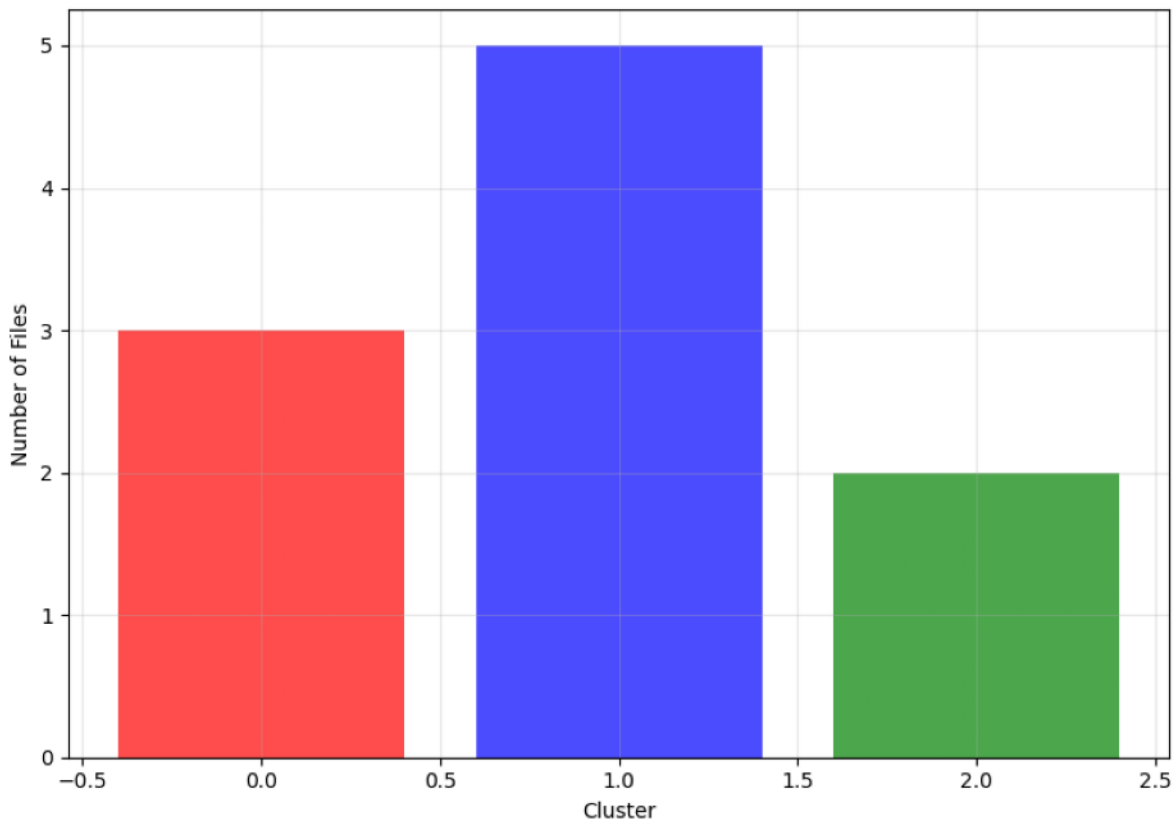
PCA - PC1 vs PC2



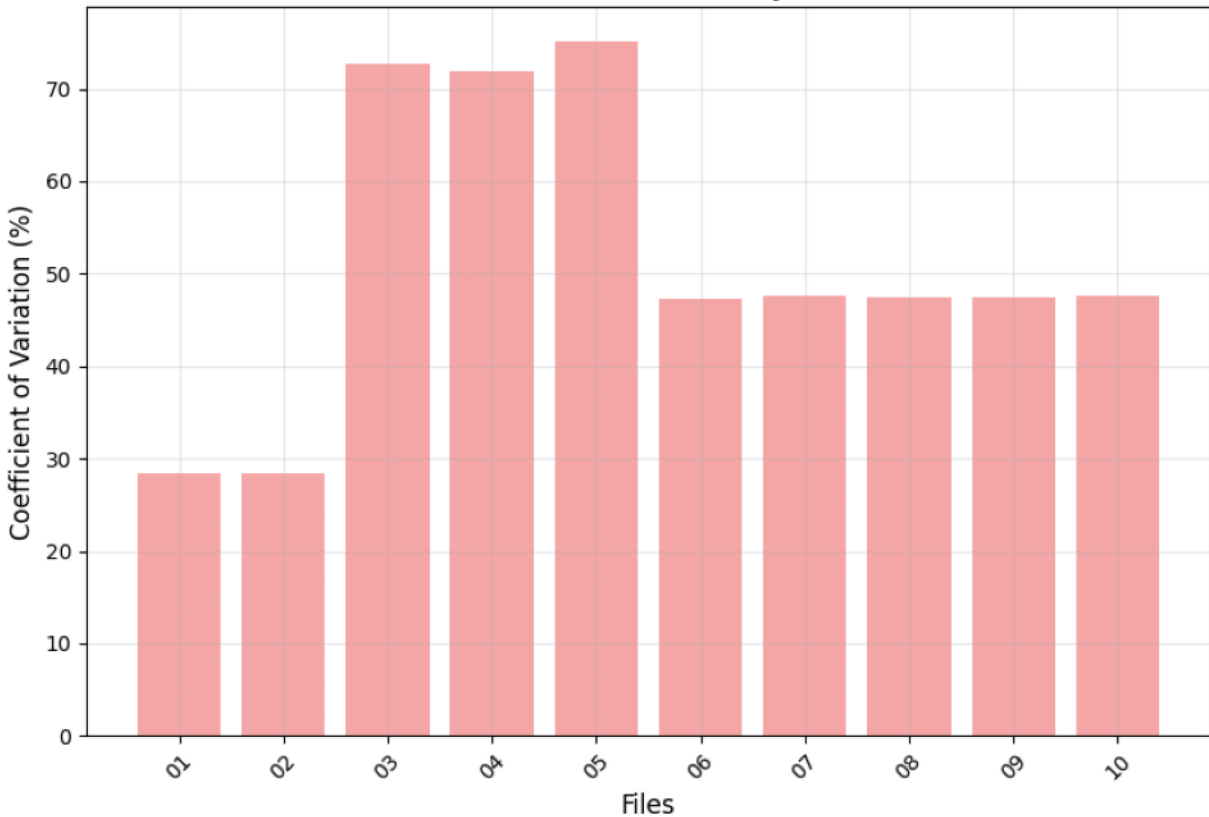
File Clustering (PCA projection)



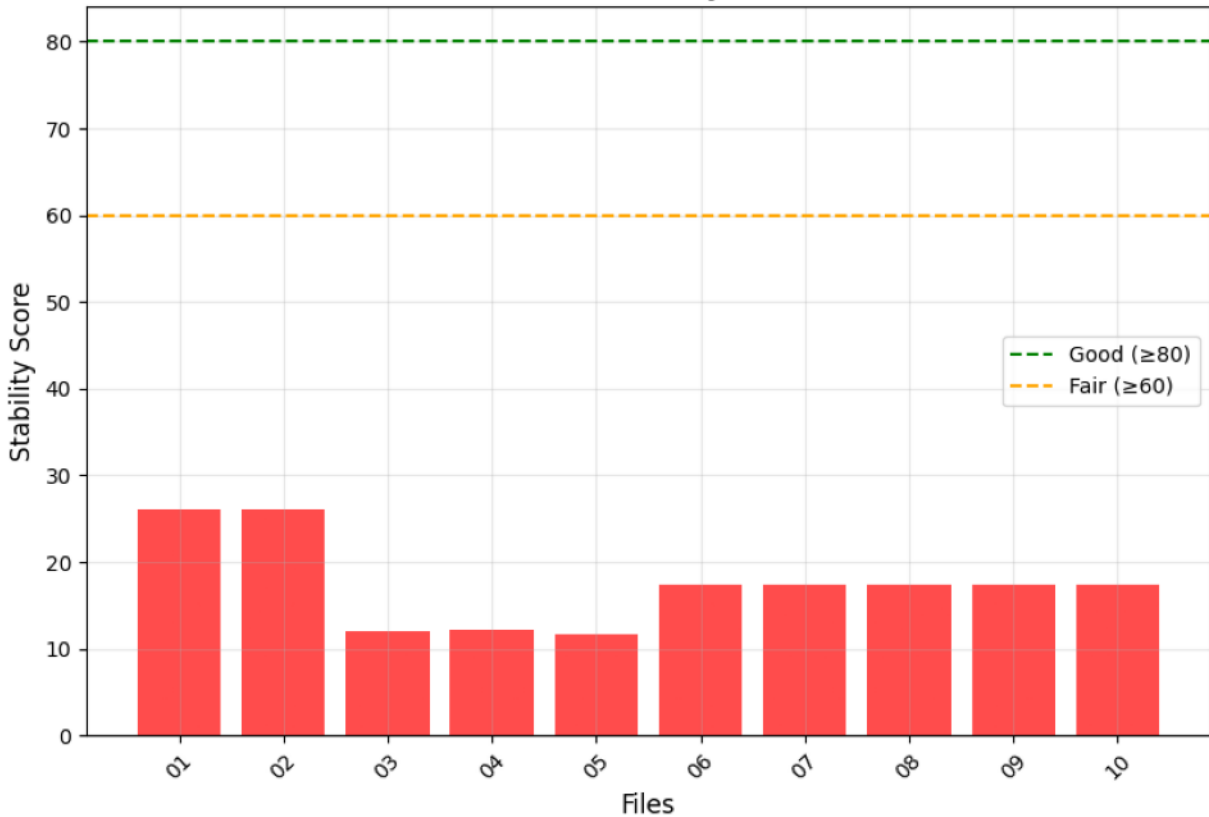
Cluster Distribution



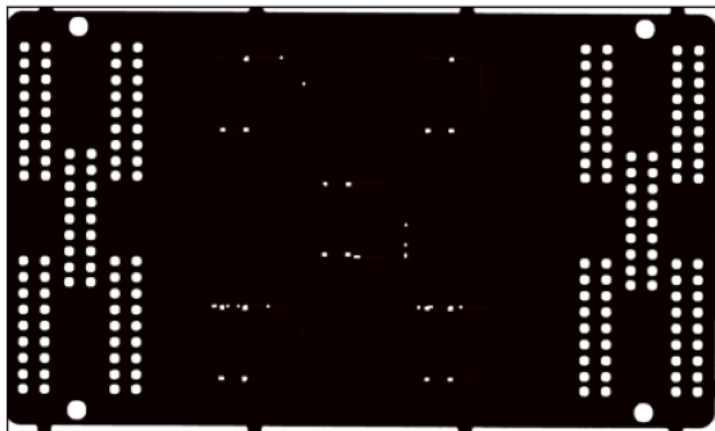
Measurement Variability (CV)



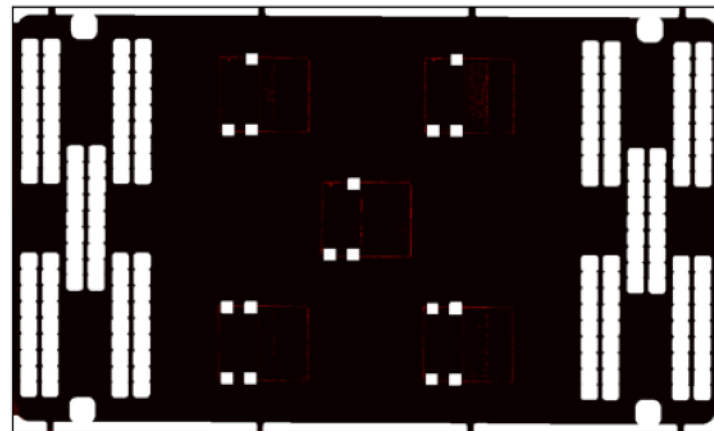
Process Stability Score



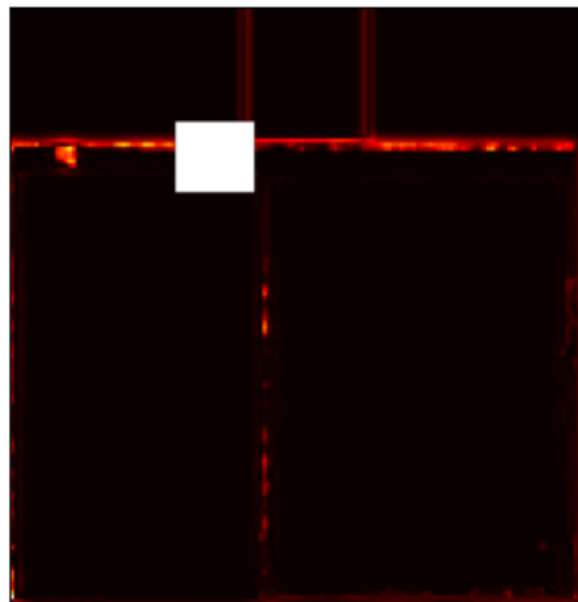
01 - Local Variability
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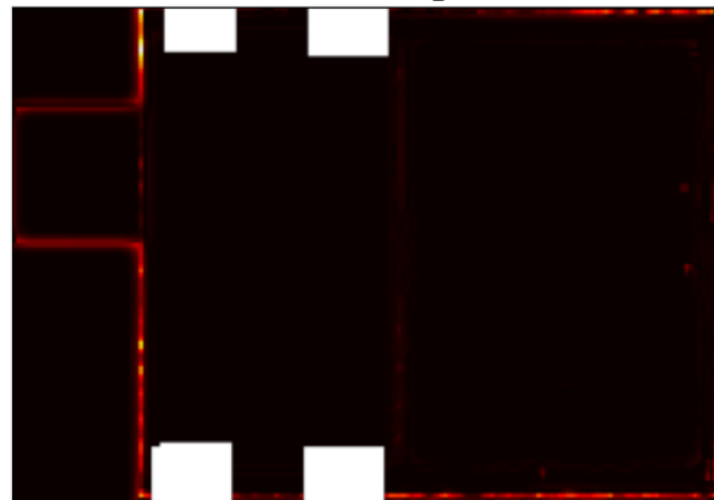
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20250716112257@_ORI.txt



03 - Local Variability
20250716112716@_ORI.txt



04 - Local Variability
20250716112912@_ORI.txt



05 - Local Variability
20250716113039@_ORI.txt



06 - Local Variability
20250716114319@_ORI.txt



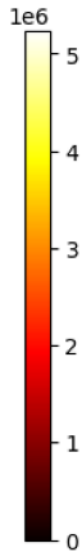
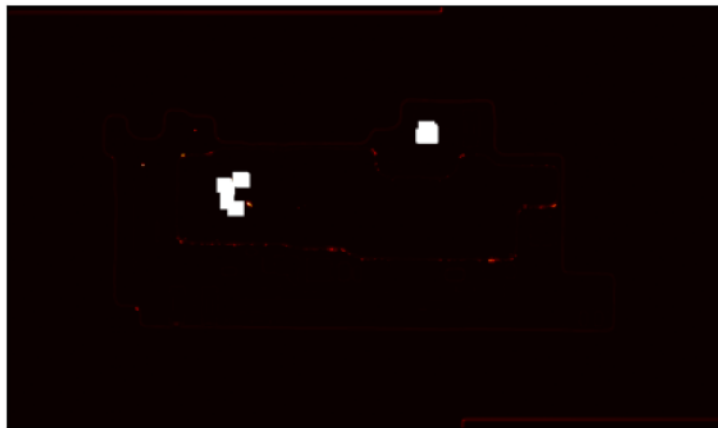
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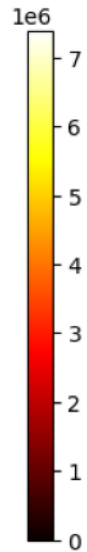
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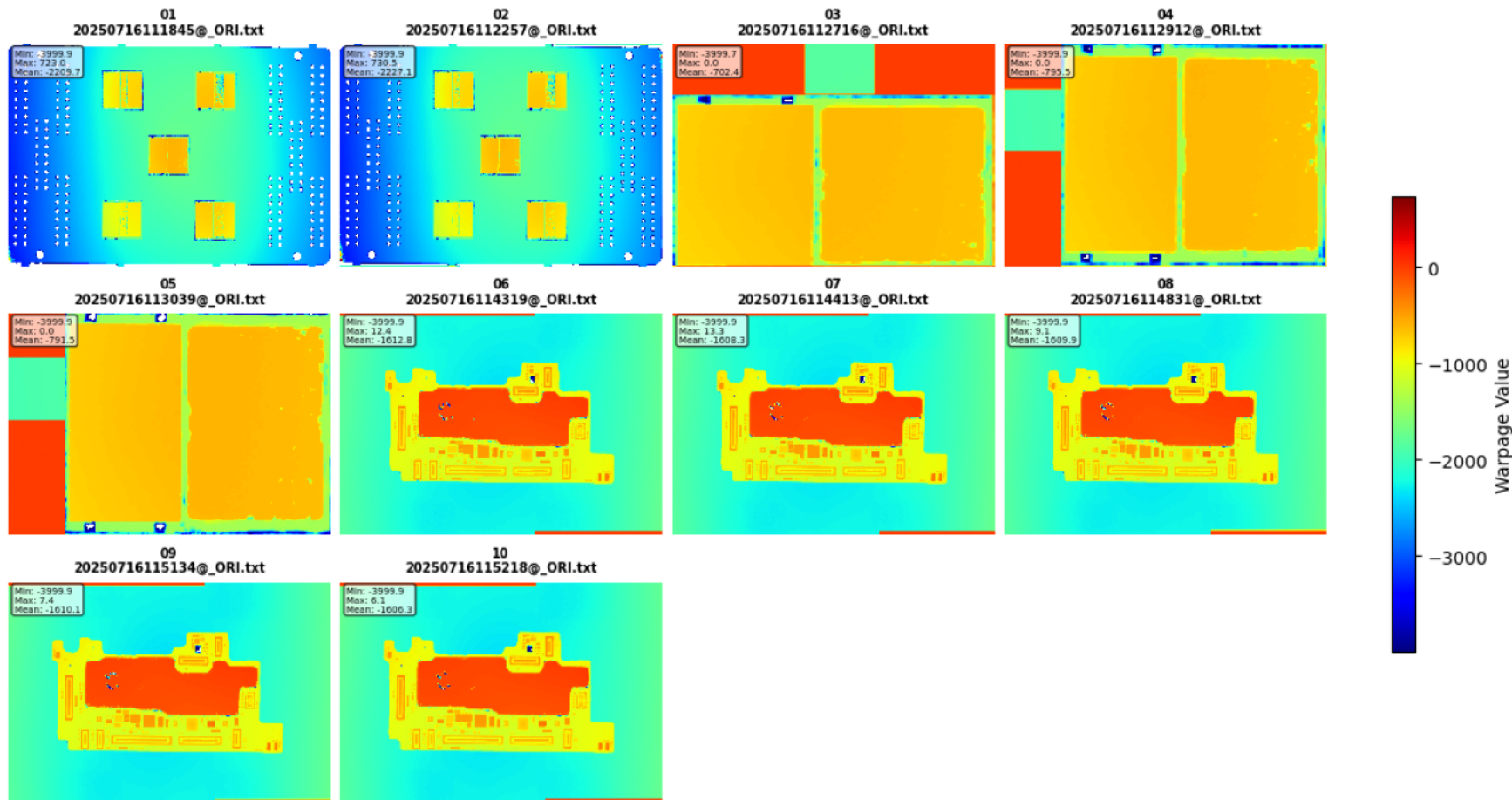
09 - Local Variability
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10 - Local Variability
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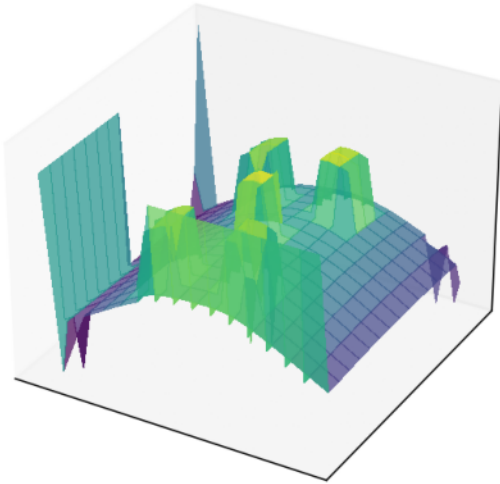


Warpage Data Comparison

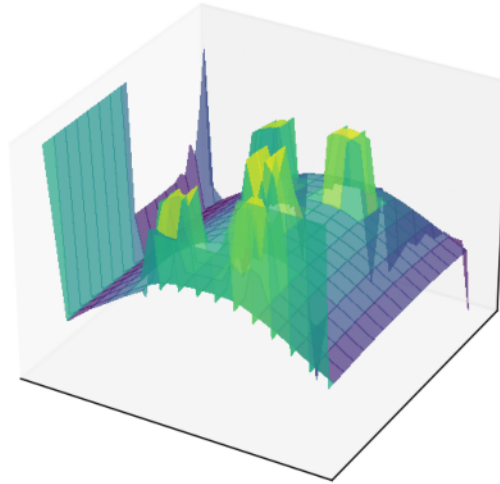


3D Surface Plots - Warpage Data (Page 1/2)

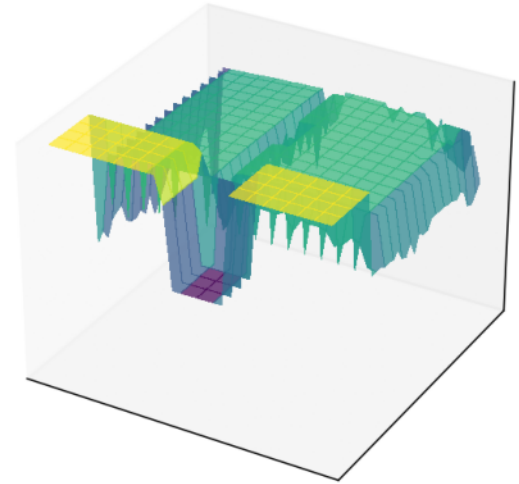
01



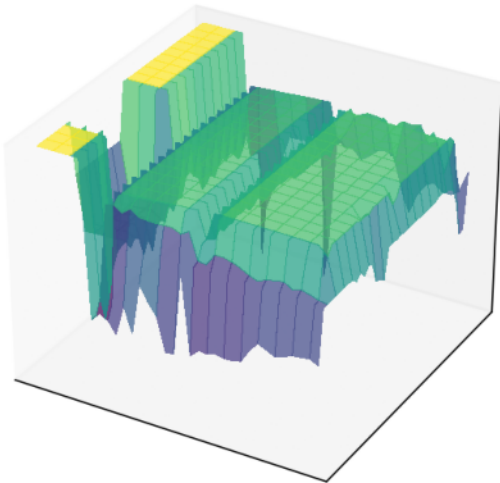
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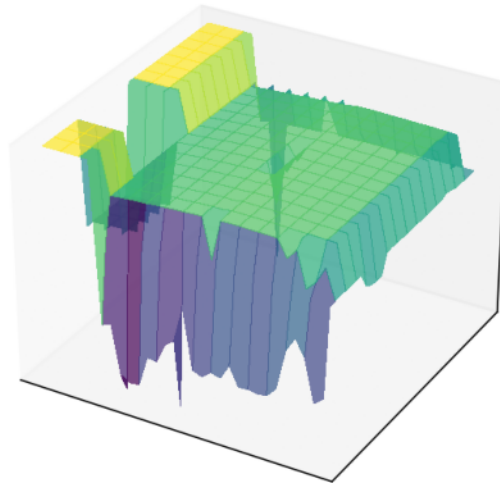
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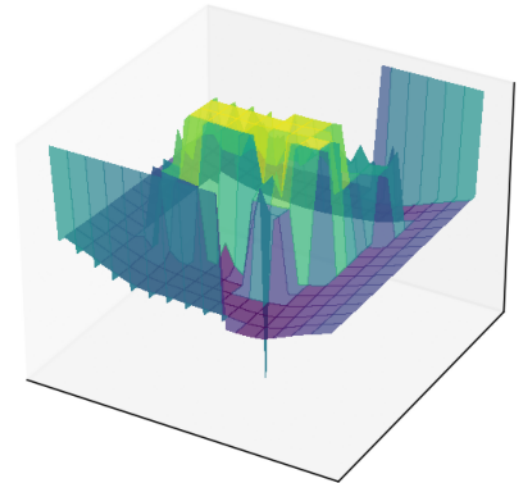
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05

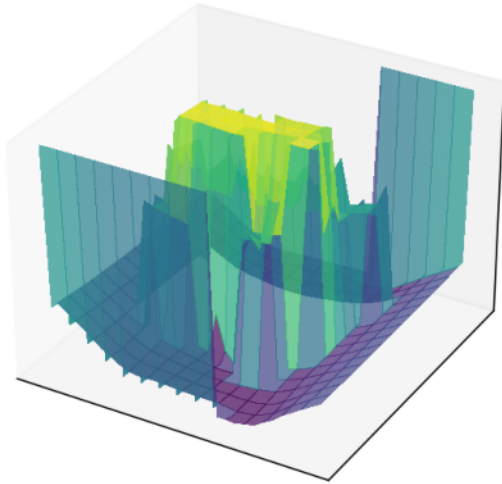


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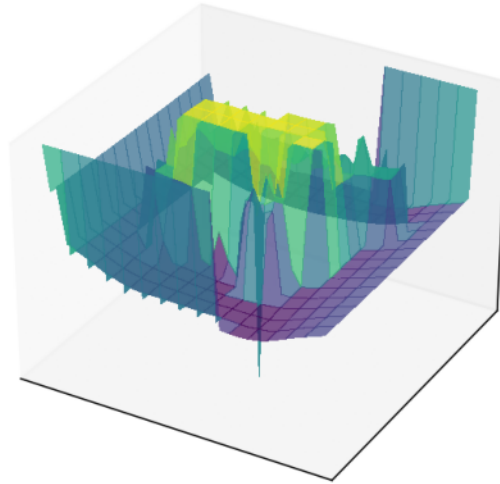


3D Surface Plots - Warpage Data (Page 2/2)

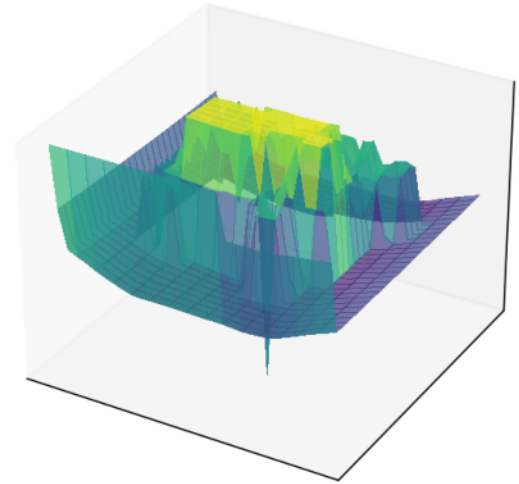
07



08



09



10

